VEVAN O NARAIN

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Third-year B.Tech CSE (AI/ML) student at VIT Vellore (9.44 GPA) seeking a Software Development Engineer internship. Strong foundation in data structures, algorithms, operating systems, and databases, with projects in backend systems, real-time collaboration, and retrieval-based applications. Focused on writing reliable, well-tested code and measuring how systems behave under load.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, C++, SQL
- **Core Concepts:** Data Structures & Algorithms, Operating Systems, DBMS, OOPs, Distributed Systems
- **Backend & Web:** Node.js, Express.js, FastAPI, REST APIs, Socket.IO, React.js, PostgreSQL, MongoDB, Redis
- **AI/ML:** Sentence Transformers, ChromaDB, RAG, BM25, HNSW, PyTorch, TensorFlow, Scikit-learn
- **Developer Tools:** Git, GitHub, Docker, VS Code, Jupyter Notebook, Linux

EDUCATION

- **B.Tech in Computer Science and Engineering (AI & ML Specialization)**
  Vellore Institute of Technology, Vellore
  Cumulative GPA: 9.44 | Expected Graduation: May, 2028
- **Senior Secondary (Class XII)** — Modern School Barakhamba Road, CBSE Board
  Percentage: 90.2% (PCM with CS) | Completed: May, 2024
- **Secondary (Class X)** — Modern School Barakhamba Road, CBSE Board
  Percentage: 96% | Completed: May, 2022

PROJECTS

Hybrid Search Engine with A/B Testing *[Python, NumPy, FastAPI, React, TypeScript, ONNX]* Demo | GitHub

- Built a hybrid search engine from scratch (HNSW index, BM25, Reciprocal Rank Fusion, int8 quantization) with no vector database, reaching 1.00 recall@10 on 16,770 chunks with a 4× memory reduction that fits a 512 MB free-tier container.
- Debugged two silent correctness bugs surfaced by benchmark anomalies: int8 quantization using only 3 of 256 levels (cosine similarity 0.71 → 0.9999) and an HNSW link-cap error leaving 41/5000 nodes reachable (recall now 0.96 at efSearch 40).
- Implemented an A/B testing framework with SHA-256 bucketing, z/Welch's t-tests, and sample ratio mismatch (SRM) detection; simulation showed a 4.3% A/A false-positive rate and 0.73 power at +2 pp.
- Developed a FastAPI backend with SSE streaming, an offline evaluation harness (recall, MRR, nDCG), and a React dashboard, backed by 36 pytest tests and a deployed live demo.

Vync - Real-Time Collaborative Editor *[React, Node.js, Socket.IO, Yjs, PostgreSQL, Redis, Docker]* GitHub

- Built a Google Docs-style collaborative code editor using Yjs CRDTs over Socket.IO, enabling low-latency concurrent editing with conflict-free synchronization across distributed clients.
- Verified convergence with a randomized fuzz harness: 10,000 concurrent edits across 20 clients produced zero divergence, with byte-identical documents on every client.
- Sandboxed code execution for 5 languages in isolated Docker containers (128 MB memory, 0.5 CPU, network disabled), and verified containment against infinite-loop, memory-bomb, fork-bomb, and outbound-network tests.
- Implemented stateless JWT/bcrypt authentication and Redis-backed room presence; measured p50 execution latency of 0.5–0.9 s for interpreted and 1.4–4.9 s for compiled languages.

10Kwery - SEC Filing RAG System *[Python, ChromaDB, Sentence Transformers, BM25, Groq, tiktoken]* GitHub

- Built an end-to-end RAG system for querying SEC 10-K/10-Q filings, returning citation-grounded answers with form-aware filtering and refusal handling for out-of-corpus queries.
- Engineered a four-channel hybrid retrieval pipeline combining dense BGE embeddings, BM25, and table-specific retrieval, merged with Reciprocal Rank Fusion (RRF) and refined by windowed Cross-Encoder reranking.
- Designed section- and table-aware preprocessing that preserves financial tables as dedicated chunks with natural-language summaries.
- Achieved 0.933 MRR, 93% Hit@1/Hit@5, and 100% citation validity on a 60-question evaluation set.

CERTIFICATIONS & ACHIEVEMENTS

- **Oracle Cloud Infrastructure Generative AI Professional** — Oracle, scored 90% (**Certificate** | **Score Report**)
- Solved 250+ problems on LeetCode across data structures and algorithms.